

Beginner Guide: Simple Todo App → Local Storage Todo App

Agar tumhe ek basic Todo App banana aata hai jisme:

- Task add hota hai
- Task delete hota hai
- Page refresh karte hi tasks gayab ho jaate hain

Toh ye guide tumhare liye hai.

Is guide me hum step-by-step ek simple Todo App ko Local Storage Todo App me convert karenge.

Starting Point

Maan lo tumhara code kuch aisa hai:

```
let inputs = document.querySelector(".input")
let text = document.querySelector(".text")

function add(){
  if(inputs.value == ""){
    alert("Please Enter Task")
  }else{

    let newEle = document.createElement("ul")
    newEle.innerHTML = `${inputs.value} <i class="fa fa-trash"></i>`

    text.appendChild(newEle)

    inputs.value = "";

    let trashItem = newEle.querySelector("i");

    trashItem.addEventListener("click", function (){
      newEle.remove();
    })
  }
}
```

Problem:

Task Add

↓

Page Refresh

↓

Sab Tasks Gayab 😭

Local Storage Kya Hai?

Browser ek chhoti storage deta hai jahan hum data save kar sakte hain.

Example:

```
localStorage.setItem("name", "Abhishek");
```

Save kar dega:

```
name = Abhishek
```

Read:

```
localStorage.getItem("name");
```

Output:

```
Abhishek
```

Important Problem

Local Storage sirf String store kar sakta hai.

Ye chalega:

```
"Hello"
```

Ye bhi:

```
"123"
```

Lekin:

```
["gym", "work", "study"]
```

Direct save nahi hoga.

Isliye:

```
JSON.stringify()
```

aur

```
JSON.parse()
```

use karte hain.

Step 1 — Tasks Array Banao

Pehle ye line add karo:

```
let tasks = [];
```

Kyu?

Abhi task sirf screen par dikh raha hai.

Kahin store nahi ho raha.

Array banega:

```
["gym", "work", "study"]
```

Step 2 — Task Ko Array Me Dalo

Else block ke andar:

```
tasks.push(inputs.value);
```

Add karo.

Example:

User likhe:

gym

Array:

`["gym"]`

User likhe:

work

Array:

`["gym", "work"]`

Step 3 — Array Ko Local Storage Me Save Karo

Add karo:

```
localStorage.setItem("myTasks", JSON.stringify(tasks));
```

Kyu?

Local Storage array nahi samajhta.

Isliye:

`["gym", "work"]`

convert hoga:

`'["gym", "work"]'`

aur save ho jayega.

Checkpoint

Browser → DevTools → Application → Local Storage

Tumhe kuch aisa dikhega:

```
Key: myTasks  
  
Value:  
["gym", "work"]
```

Agar ye dikh raha hai toh tum sahi ja rahe ho.

Step 4 — Local Storage Se Data Nikalna

Purani line:

```
let tasks = [];
```

Remove karo.

Uski jagah:

```
let tasks = JSON.parse(localStorage.getItem("myTasks")) || [];
```

Lagao.

Ye Line Kya Kar Rahi Hai?

Part 1:

```
localStorage.getItem("myTasks")
```

Storage se value nikalti hai.

Part 2:

```
JSON.parse()
```

String ko wapas array banata hai.

Example:

```
'["gym","work"]'
```

↓

```
["gym","work"]
```

Part 3:

```
|| []
```

Agar storage empty ho toh empty array use karo.

Step 5 — Sabse Bada Refactor

Yahan bahut log confuse hote hain.

Main bhi yahin confuse hua tha.

Question:

Page refresh ke baad tasks array me aa gaye.

Lekin screen par kaise dikhenge?

Answer:

Task banane wala code alag function me nikalna padega.

renderTask() Function Banao

```
function renderTask(taskText){  
  
}
```

Yahan Main Confuse Hua Tha

Mujhe laga tha:

```
Else ke sabhi lines renderTask me chali jayengi
```

Lekin ye galat hai.

Data Management Aur UI Alag Hote Hain

Ye lines:

```
tasks.push(inputs.value);  
  
localStorage.setItem(...);
```

Data management hain.

Ye add() me rahengi.

Ye lines:

```
let newEle = document.createElement(...)  
  
newEle.innerHTML = ...  
  
text.appendChild(...)  
  
trashItem.addEventListener(...)
```

UI rendering hain.

Ye renderTask() me jayengi.

Important Confusion

Pehle:

```
newEle.innerHTML = `${inputs.value}`
```

Tha.

Ab:

```
newEle.innerHTML = `${taskText}`
```

Hoga.

Kyu?

Kyuki refresh ke baad input box khali hota hai.

Tab hume array se milne wale task ko show karna hai.

Common Mistake

Galat:

```
inputs.value = "";  
renderTask(inputs.value);
```

Kyu galat?

Pehle input clear ho gaya.

Ab:

```
inputs.value
```

=

```
""
```

Matlab empty task render hoga.

Sahi Order

```
renderTask(inputs.value);  
  
inputs.value = "";
```


Pehle render.

Fir input clear.

Step 6 — Page Load Par Tasks Show Karna

renderTask function ke niche:

```
tasks.forEach(task => renderTask(task));
```

Add karo.

Ye Kya Karega?

Agar:

```
tasks = ["gym", "work", "study"]
```

Hai.

Toh:

```
renderTask("gym")  
renderTask("work")  
renderTask("study")
```

Call hoga.

Aur saare tasks screen par aa jayenge.

Step 7 — Delete Bug

Ab problem:

Delete karte ho.

Refresh karte ho.

Task wapas aa jata hai.

Kyu?

Kyuki:

```
newEle.remove();
```

Sirf screen se hata raha hai.

Local Storage se nahi.

Delete Ka Correct Flow

```
Trash Click  
↓  
Array Se Remove  
↓  
Local Storage Update  
↓  
Screen Se Remove
```

filter() Samajho

Line:

```
tasks = tasks.filter(task => task !== taskText);
```

Shuru me mujhe bhi confusing lagi thi.

Example:

```
tasks = ["gym", "work", "study"]
```

Delete:

```
taskText = "work"
```

Filter check karega:

```
gym !== work → true → rakho  
work !== work → false → hatao  
study !== work → true → rakho
```

Result:

```
["gym", "study"]
```

Fir Local Storage Update

```
localStorage.setItem("myTasks", JSON.stringify(tasks));
```

Fir refresh ke baad deleted task wapas nahi aayega.

Final Delete Code

```
trashItem.addEventListener("click", function () {  
    tasks = tasks.filter(task => task !== taskText);  
    localStorage.setItem("myTasks", JSON.stringify(tasks));  
    newEle.remove();  
})
```

Important Limitation

Agar:

```
gym  
work  
gym
```

Add kiya.

Aur ek gym delete kiya.

Toh dono gym delete ho jayenge.

Kyu?

Kyuki hum text ke basis par delete kar rahe hain.

Professional apps me unique ID use hoti hai.

Example:

```
{  
  id: 1,  
  text: "gym"  
}
```

Ye advanced topic hai.

Tumne Is Project Me Kya Seekha?

- ✓ DOM Manipulation
 - ✓ createElement()
 - ✓ querySelector()
 - ✓ addEventListener()
 - ✓ Arrays
 - ✓ push()
 - ✓ JSON.stringify()
 - ✓ JSON.parse()
 - ✓ localStorage.setItem()
 - ✓ localStorage.getItem()
 - ✓ filter()
 - ✓ forEach()
 - ✓ Function Refactoring
 - ✓ Data vs UI Separation
-

Final Achievement

Ab tumhara Todo App:

- ✓ Add Task
- ✓ Delete Task
- ✓ Save In Local Storage
- ✓ Load After Refresh
- ✓ Dynamic Rendering

Support karta hai.

Ye beginner JavaScript ka ek solid project hai.